TUTORIAL: Digital Out

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I. Introduction

In this lab, we will learn how to control the Digital Output of GPIOs of the MCU board to turn on/off an LED.

The objectives of this lab are to learn how to

- Read and configure registers of digital GPIO of MCU
- Program firmware to control digital input/output pins

Hardware

NUCLEO -F411RE

Software

Keil uVision IDE, CMSIS, EC_HAL

Documentation

STM32 Reference Manual

II. Basics of GPIO Out

A. Bit Operation

Fill in the blanks. You should write the answer in both hexa-decimal and binary number.

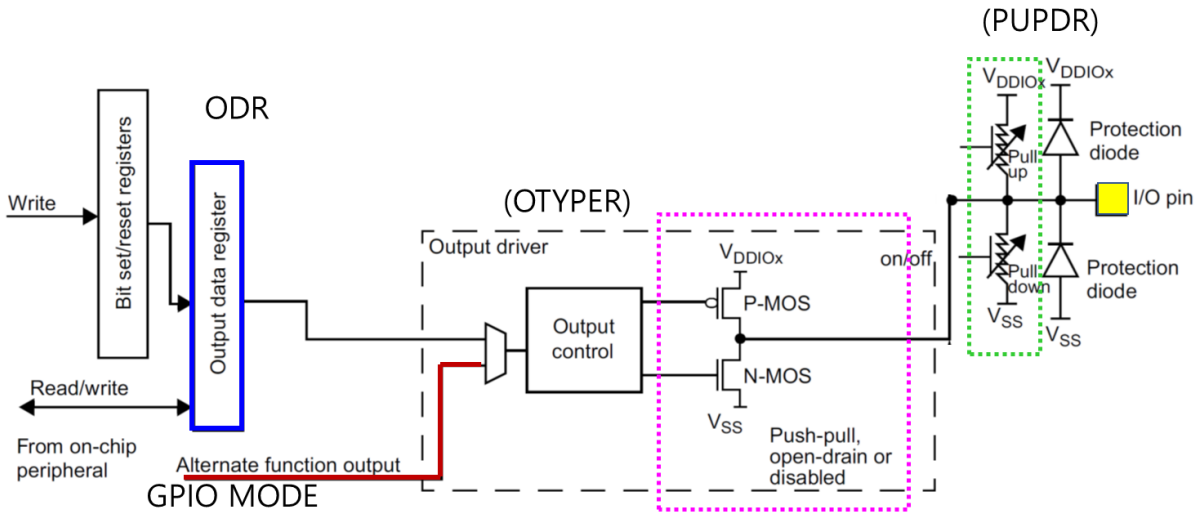
Bit operation	Description	Answer
0xA & 0x2	AND	0x2, b10
0x3 | 0xC	OR	0xF, b1111
~0x7	NOT	0x8, b1000
0xF ^ 0x5	XOR(Toggle)	0xA, b1010
0x1 << 3	Shift left	0x8, b1000
0xC >> 2	Shift right	0x3, b11
0x11 |= 1<<3	Set bit	0x19, b11001
0xFF &= ~1<<4	Clear bit	0xEF, b11101111
val=0x0F>>3 & 1	Read bit	0x01, b1

B. GPIO Digital Out Register

List GPIO registers for this LAB

Type	Register Name	Description
GPIO	GPIOx_MODER	Mode: Output/Input/Analog/Alternate function
	GPIOx_OTYPER	Output Type: Opendrain/Push-Pull
	GPIOx_OSPEEDR	Output Speed: LOW/Medium/Fast/High
	GPIOx_PUPDR	Pull-Up Pull-Down: No pull-up,down/Pull-up, Pull-down, Reserved
	GPIOx_ODR	Output Data Register

Schematic



Process of GPIOx register initiation

0. Enable Peripheral Clock (**AHB1ENR**)
1. Configure as Digital Output (**MODER**)
2. Configure pull-up/down resistors (**PUPDR**)
3. For Output: Configure Output Type (**OTYPE**)
4. For Output: Configure Output Speed (**OSPEEDR**)

5. Output Data **(ODR)**

III. Tutorial

A. Register Configuration

1. GPIO: Digital Out - Pin Initialization & Set LED

Port A Pin 5 / Output / Push-Pull / No Pull-Up Pull-Down / Default Speed

use **#define LED_PIN 5**

- **MODER:** Output (MODER5[1:0]=01)

```
GPIOA->MODER & = ~(3<<( LED_PIN *2));     // clear bits at both  [10] and [11]
GPIOA->MODER | = 1<<(LED_PIN *2);         // set bit at [10]
```

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **MODERy[1:0]:** Port x configuration bits (y = 0..15)
These bits are written by software to configure the I/O direction mode.
00: Input (reset state)
01: General purpose output mode
10: Alternate function mode
11: Analog mode

Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Initial	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Logic	Bitwise OR																															
mask	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
Value	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	1	x	x	x	x	x	x	x	x	x	x

- **OTYPER:** Push-Pull (OT5=0)

```
GPIOA->OTYPER &= ~(1<< LED_PIN);
```

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **OTy**: Port x configuration bits (y = 0..15)
These bits are written by software to configure the output type of the I/O port.
0: Output push-pull (reset state)
1: Output open-drain

Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Initial																	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Logic	Bitwise AND																															
mask	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
Value	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x

- **PUPDR:** no pull-up pull-down (PUPDR5[1:0]=00)

```
GPIOA->PUPDR &=   ~(3<<( LED_PIN *2));      // clear
GPIOA->PUPDR |=   (0UL<<(LED_PIN*2));        //set
```

Paste Register map from reference manual

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PUPDR15[1:0]		PUPDR14[1:0]		PUPDR13[1:0]		PUPDR12[1:0]		PUPDR11[1:0]		PUPDR10[1:0]		PUPDR9[1:0]		PUPDR8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PUPDR7[1:0]		PUPDR6[1:0]		PUPDR5[1:0]		PUPDR4[1:0]		PUPDR3[1:0]		PUPDR2[1:0]		PUPDR1[1:0]		PUPDR0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **PUPDRy[1:0]:** Port x configuration bits (y = 0..15)
These bits are written by software to configure the I/O pull-up or pull-down
00: No pull-up, pull-down
01: Pull-up
10: Pull-down
11: Reserved

- **ODR:** Set LED (ODR5=1)

```
GPIOA->ODR |= (1UL<<LED_PIN);
```

Paste Register map from reference manual

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODRy**: Port output data (y = 0..15)
These bits can be read and written by software.

Note: For atomic bit set/reset, the ODR bits can be individually set and reset by writing to the GPIOx_BSRR register (x = A..E and H).

B. Programming

Preparation

- Download ecRCC.h and ecRCC.c.
- Save them in the library folder: "**include\"**
  - ➢ You must have "ecPinNames.*, ecRCC.*" in include folder
- Create a new item(file) as '**\tutorial\TU_GPIO_Digital_Out_LED.c'**
- Copy and paste from the source code '**TU_GPIO_Digital_Out_LED_student.c**'.
- Create a new environment : '**TU_GPIO_Digital_Out_LED**'.

(platformio.ini) : `[env:TU_GPIO_Digital_Out_LED]`

```
[env:TU_GPIO_Digital_Out_LED]
build_src_filter = +<tutorial/TU_GPIO_Digital_Out_LED.c> +<include/*.c> -
<include/ecSTM32_simple.c> (not needed since it was already deleted!)
```

  - ➢ Here, we have removed ecSTM32_simple.h for it can cause redundant errors
  - ➢ You can also manually delete "ecSTM32_simple.*" files.

Exercise

This is an example code of turning ON LED *without* button input. Do not worry if you do not understand what this code means yet. You will learn one by one in the following few weeks.

Fill in the empty spaces in the code.

Then, compile and flash the source code onto the MCU board.

Verify the program by checking if the LED is turned ON.

Solution

```c
#include "stm32f411xe.h"
#include "ecRCC2.h"

#define LED_PIN    PA_5     //LD2

int main(void) {
    /* Part 1. RCC GPIOA Register Setting */
    RCC_HSI_init();
    RCC_GPIOA_enable();

    /* Part 2. GPIO Register Setting */
    // GPIO Mode Register : MODE=OUTPUT
    GPIOA->MODER &= ~(3UL<<(2*LED_PIN));    // Clear '00' for Pin 5
    GPIOA->MODER |=   1UL<<(2*LED_PIN);     // Set '01' for Pin 5

    // GPIO Output Type Register
    GPIOA->OTYPER &= ~(1UL<<LED_PIN);       // 0:Push-Pull

    // GPIO Pull-Up/Pull-Down Register
    GPIOA->PUPDR  &= ~(3UL<<(2*LED_PIN));   // 00: none

    // GPIO Output Speed Register
    GPIOA->OSPEEDR &= ~(3UL<<(2*LED_PIN));
    GPIOA->OSPEEDR |=   2UL<<(2*LED_PIN);   //10:Fast Speed

    // Dead loop & program hangs here
    while(1){
        GPIOA->ODR |= (1UL << LED_PIN);     // Change only LED_PIN = H
    }
}
```

Appendix

[See here for MCU resources]

1. Pin Configuration of NUCLE-F401RE

Figure 18. NUCLEO-F401RE

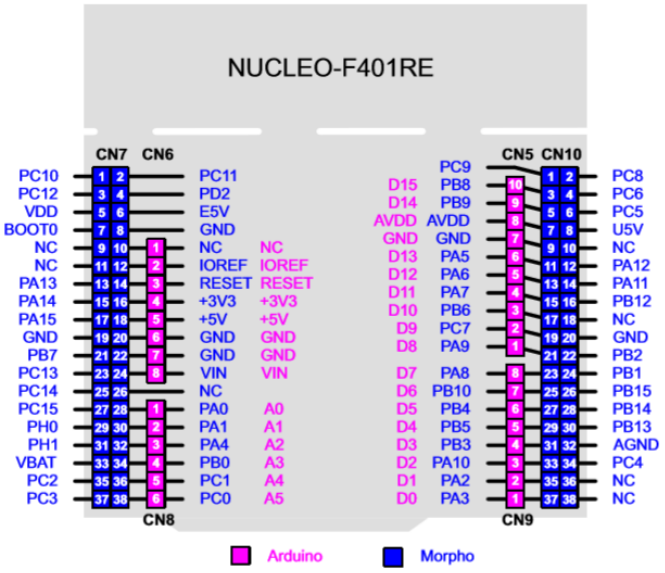


Table 29. ST morpho connector on NUCLEO-F401RE, NUCLEO-F411RE, NUCLEO-F446RE

CN7 odd pins		CN7 even pins		CN10 odd pins		CN10 even pins	
Pin	Name	Name	Pin	Pin	Name	Name	Pin
1	PC10	PC11	2	1	PC9	PC8	2
3	PC12	PD2	4	3	PB8	PC6	4
5	VDD	E5V	6	5	PB9	PC5	6
7	BOOT0(1)	GND	8	7	AVDD	U5V(2)	8
9	-	-	10	9	GND	-	10
11	-	IOREF	12	11	PA5	PA12	12
13	PA13(3)	RESET	14	13	PA6	PA11	14
15	PA14(3)	+3.3V	16	15	PA7	PB12	16
17	PA15	+5V	18	17	PB6	-	18
19	GND	GND	20	19	PC7	GND	20
21	PB7	GND	22	21	PA9	PB2	22
23	PC13	VIN	24	23	PA8	PB1	24
25	PC14	-	26	25	PB10	PB15	26
27	PC15	PA0	28	27	PB4	PB14	28
29	PH0	PA1	30	29	PB5	PB13	30
31	PH1	PA4	32	31	PB3	AGND	32
33	VBAT	PB0	34	33	PA10	PC4	34
35	PC2	PC1 or PB9(4)	36	35	PA2	-	36
37	PC3	PC0 or PB8(4)	38	37	PA3	-	38

1. Default state of BOOT0 is 0. It can be set to 1 when a jumper is on pin5-7 of CN7.Two unused jumpers are available on CN11 and CN12 (bottom side of the board).
2. U5V is 5 V power from ST-LINK/V2-1 USB connector and it rises before +5V.
3. PA13 and PA14 share with SWD signals connected to ST-LINK/V2-1, it is not recommend to use them as IO pins if ST-LINK part is not cut.
4. Refer to *Table 10: Solder bridges* for details.

2. LED/Button Circuit Diagram

